A Mechanistic Model to Weigh the contributions of Trisomy 21 in Atrioventricular Septal Defect Disorders

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Abstract

Atrioventricular septal defects (AVSDs) are highly enriched in trisomy 21 (Down syndrome), yet exhibit incomplete penetrance, suggesting a multi-factorial developmental failure process. Existing genetic and developmental studies support a threshold-based interpretation of disease risk, but no integrated dynamical model links genotype to morphogenetic failure. Here, we develop a mechanistically grounded dynamical systems model of AV septal development incorporating endocardial cushion growth, dorsal mesenchymal protrusion dynamics, extracellular matrix remodeling, and hemodynamic feedback. Genetic perturbations are modeled as parameter shifts in biologically interpretable axes. We define both linear and nonlinear phenotype maps and introduce an explicit geometric closure criterion. This framework provides a quantitative basis for understanding AVSD as a threshold-crossing process and generates testable predictions linking genetic burden, developmental timing, and biomechanical environment.

1 Introduction

1.1 Background

Atrioventricular septal defects (AVSDs) comprise a range of different disorders, all originating from the partial to complete absence of the atrioventricular septation. While the exact causes of the defect are unknown, it has been especially implicated in children born with Trisomy 21, or Down Syndrome (DS). Reported incidence in DS ranges, but it is estimated that between 15 and 33% are born with AVSD. Moreover, we still do not have an explanation for why AVSD prevalence arises in DS, as well as the morphology in development. There is some established research on the general links between DS and congenital heart defects (CHD), with the prevalence of any defect in any given infant with DS at about 40-50%. While patterns of disease vary throughout different geographical populations, most likely due to socio-demographic factors, profiles have been made for different global regions. As for mechanistic understanding between the two, we do not have a complete understanding. However, researchers have proposed several candidate genes linked in the increased risk of CHD: DSCAM, COL6A1, COL6A2, KCNJ6 and RCAN1. DSCAM has particularly been implicated in AVSD, as the gene located on 21q22.2 is involved in cell adhesion. Barlow et al. proposed that the abnormal expression of intercelluar mucoproteins linked to DSCAM affects endocardial fusion, thus causing AVSD. Before diving into the complete pathogenesis of AVSD, it is important to gain both an understanding in atrioventricular structure and a distinction between the different subtypes. In a typical heart, newly oxygenated blood is held in the right atrium and ventricle. It is separated from the deoxygenated blood in the left atrium and ventricle. Separating the atrium and ventricles, are the interatrial and interventricular septums, respectively. In AVSD, the barrier between the left and right side of the hearts can be absent, and thus deoxygenated and oxygenated blood will mix. The level of barrier missing can vary as well. If only part of the septum

is asbent, it is partial AVSD. Usually partial AVSD will present with the interatrial space missing. However, in complete AVSD both the interatrial and interventricular space may not develop.

1.2 AVSD Pathogenesis

To reiterate, the exact pathogenesis of AVSD are blurry, but what has long been accepted is that precursor tissue for the atrioventricular (AV) valve leaflets was to blame for the development of AVSD. These tissues are usually referred to as the endocardial AV cushions (EAVC). However, others have also proposed that the abnormal development of the dorsal mesenchymal protusion (DMP) was responsible for the defects. Nonetheless, EAVC and DMP work together in embryologic development to construct the atrial septum, and thus the contribution of each remains unclear. What we do know is that the endocardial EAVCs arise through endocardial-to-mesenchymal transition (EndMT), beginning at approximately embryonic day (E)9.5 in mouse models, and expand through both continued mesenchymal influx and local proliferation. These cushions form the primary structural foundation for AV septation and valve formation, and defects in their size, composition, or remodeling have long been associated with AVSD.

Concurrently, cells from the posterior second heart field contribute to the formation of the DMP, which extends into the AV canal and interacts with the developing cushions. The timing of DMP deployment is restricted to a narrow developmental window (approximately E9.5–E10.5 in mouse), and proper septation depends on the coordinated interaction between DMP-derived tissue and cushion mesenchyme. Disruptions in either the magnitude or timing of these contributions can impair septation, even when neither structure is completely absent.

In addition to cellular contributions, the biomechanical environment plays a critical role in shaping AV development. As cardiac function initiates and blood flow increases, hemodynamic forces influence both cushion remodeling and extracellular matrix (ECM) organization. Shear stress and pressure-driven cues regulate matrix deposition, tissue compaction, and structural maturation, meaning that septation also depends on the mechanical environment in which these tissues develop.

Importantly, these processes occur within finite developmental windows, after which the opportunity for proper septation diminishes. As a result, delays or modest reductions in cushion formation, DMP contribution, or ECM maturation may not be fully compensated if they occur outside critical time intervals. This introduces the possibility that AVSD arises not from the complete failure of a single developmental pathway, but from the cumulative effects of multiple small perturbations that prevent the system from reaching a necessary structural threshold within the available developmental timeframe.

1.3 Contribution of this work

While much extensive work has been done to narrow down the individual causes of AVSD and it's link to DS, no one has taken a mathematical and data driven approach towards the pathologies. With the possibility that AVSD penetrance follows a combination of multiple different factors, I propose that one can create a probabilistic model to weigh the individual contributions of each factor. In this work, I present a threshold-based model of AV septation that integrates key developmental processes, including EAVC growth, DMP contribution, ECM remodeling, and hemodynamic effects. Genetic perturbations associated with trisomy 21 are incorporated as parameter shifts within biologically interpretable axes, allowing the model to bridge genotype and developmental dynamics.

Importantly, this framework enables direct comparison between additive (linear) and interaction-based (nonlinear) representations of developmental sufficiency, providing a means to test whether

AVSD arises from independently accumulating deficits or from coupled failure events across multiple systems. In addition, the model supports quantitative analysis of subsystem contributions through sensitivity and perturbation-based approaches, allowing identification of which developmental processes most strongly influence AVSD risk.

Rather than attributing AVSD to a single dominant defect, this approach addresses the hypothesis that disease emerges when the combined effects of multiple small perturbations prevent the developing heart from achieving sufficient structural closure. This provides a quantitative framework for understanding incomplete penetrance in Down syndrome and generates testable predictions linking genetic variation, development, and biomechanical environment.

1.4 Probability Threshold Model

It is important that we gain a basic understanding of threshold models before diving in. Threshold models describe systems in which an outcome is determined by whether an underlying continuous variable exceeds a critical value. Let X denote a latent variable representing the combined effect of multiple contributing factors. A simple deterministic threshold model defines the phenotype as

$$Y = \begin{cases} 1, & X \geq \tau \\ 0, & X < \tau \end{cases}$$

where τ is a critical threshold and Y denotes the presence or absence of a phenotype.

In many biological systems, X is not directly observable and arises from the combined influence of multiple variables. A common formulation expresses X as a function of contributing components:

$$X = f(x_1, x_2, \ldots, x_n)$$

where each x_i represents a biological factor. In the simplest case, this combination is additive,

$$X = \sum_{i=1}^{n} w_i x_i,$$

while more complex models allow for nonlinear interactions,

$$X = \prod_{i=1}^{n} x_i^{w_i} \quad \text{or} \quad \log X = \sum_{i=1}^{n} w_i \log x_i.$$

To account for variability between individuals, threshold models are often extended to probabilistic formulations. Let X be treated as a random variable with distribution $p(X)$. The probability of observing the phenotype is then

$$P(Y = 1) = P(X \geq \tau).$$

In practice, this is often approximated using a smooth function such as a logistic mapping:

$$P(Y = 1) = \frac{1}{1 + \exp\left(-k(X - \tau)\right)},$$

where k controls the sharpness of the transition.

In developmental systems, the latent variable X can be interpreted as a measure of functional or structural sufficiency accumulated over time. In this setting, X may depend on time-dependent processes, and the phenotype is determined by whether a sufficient level is reached by a critical developmental time T:

$$Y = \begin{cases} 1, & X(T) \geq \tau \\ 0, & X(T) < \tau. \end{cases}$$

This formulation naturally captures scenarios in which multiple small perturbations combine to produce failure, particularly when the system operates within a finite temporal window.

2 Methods

2.1 State variables and normalization

In order to construct a dynamical model out of AVSD, we must first consider where biological components are most important to a successful septal formation. As a reminder, we can consider septal formation as a function of EAVC growth, DMP contributions, ECM remodeling, and hemodynamic effects. Modeling individual groups of cells or using rigorous fluid dynamics would exceed our computational capacity. Thus, we represent each of these processes through continuous state variables that capture their effective contribution to septation, rather than setting up a large number of individual contributions. This allows us to describe the system at a level that is both biologically interpretable and mathematically reasonable.

To start, our independent variable, t, will denote developmental time in days, where $t = 0$ is at E9.0. Then, let us define a state variable for endocardial cushion mesenchymal mass, $C(t)$. This variable represents the effective amount of mesenchymal tissue contributed by the EAVCs. It is included because the cushions form the primary structural scaffold for AV septation and valve formation, and abnormalities in cushion growth, cellularization, or remodeling have long been implicated in AVSD pathogenesis.

Second, we define a state function $D(t)$ to represent the contribution of DMP tissues. As a reminder, DMP arises from the posterior second heart field and contributes mesenchymal tissue that integrates with the developing cushions and is essential for proper atrioventricular septation.

Then, we define our geometric gap. This variable represents the remaining unclosed portion of the atrioventricular septal region. It is included to provide a direct structural readout of septation progress. Whereas the other state variables describe the biological subsystems that promote closure, $G(t)$ represents the anatomical consequence of their combined action and serves as the most physically interpretable criterion for whether septation ultimately succeeds or fails. It is important to note that $C(t)$, $D(t)$, and $G(t)$ are extensive quantities, as they are a count of cells or mass. For simplicity, we normalize the two according to the following:

$$c(t) = \frac{C(t)}{C^*}, \qquad d(t) = \frac{D(t)}{D^*}, \qquad g(t) = \frac{G(t)}{G^*}$$

where C^* and D^* denote reference values corresponding to sufficient tissue contributions under normal development. However, G^* denotes the initial unclosed septal gap.

Unlike the previous state variables, the two following variables are naturally normalized. This is because they are naturally intensive and represent a fraction similar to quality, rather than quantity. The first of these variables I will define is $E(t)$, a measure of ECM maturation. This variable represents the degree of ECM organization and remodeling. It is included because septation depends not only on the presence of tissue, but also on the structural environment in which that tissue develops.

For the last of our state variables, I will define $A(t)$, the level of AV canal competence. It is included to capture the inductive signaling and regional myocardial identity required to support

proper cushion and DMP development. Rather than modeling individual signaling pathways explicitly, this variable serves as a coarse-grained measure of whether the local developmental program is favorable for normal septation.

To summarize, I have defined five state variables that will be essential to defining the probabilistic model later to come. For a detailed overview of each variable, see Table 1.

Variable	**Description**	**Type**	**Range**
t	Developmental time (days, $t = 0$ at E9.0)	Independent	$t \geq 0$
$C(t)$	Endocardial cushion mesenchymal mass	Extensive	$C(t) \geq 0$
$D(t)$	Dorsal mesenchymal protrusion (DMP) contribution	Extensive	$D(t) \geq 0$
$G(t)$	Geometric septation gap (unclosed region)	Extensive	$G(t) \geq 0$
$E(t)$	ECM maturation state	Intensive	$[0, 1]$
$A(t)$	AV canal competence	Intensive	$[0, 1]$

Table 1: Summary of state variables used in the AVSD dynamical model.

2.2 Developmental timing windows

In AVSD, we are hypothesizing that multiple morphogenetic events and processes must occur in specific developmental windows. In particular, endocardial-to-mesenchymal transition (EndMT) giving rise to the atrioventricular cushions begins around embryonic day (E)9.5 in the mouse, while dorsal mesenchymal protrusion (DMP) contribution from the posterior second heart field occurs over a similarly restricted interval between approximately E9.5 and E10.5. Subsequent remodeling and fusion events occur later, with major cushion fusion occurring near E12.5.

Rather than modeling abrupt on/off behavior, we introduce smooth timing functions that modulate the activation of each developmental subsystem. For convenience, we represent each window using a sigmoidal gating function:

$$W(t; t_{\text{on}}, t_{\text{off}}, \Delta) = \sigma\left(\frac{t - t_{\text{on}}}{\Delta}\right) \sigma\left(\frac{t_{\text{off}} - t}{\Delta}\right), \qquad \sigma(x) = \frac{1}{1 + e^{-x}}$$

The reasoning behind this modeling is simple; We need to recapitulate a biological process where we can model a dynamical system that is time limited, smooth, and can be turned on or off simply. The use of two sigmoid functions allows for simple on and off switch, while the sigmoid function itself gives a controllable yet smooth transition in our time space.

Furthermore, we know that most issues arise earlier in development. Thus, we will need a way to penalize later developmental issues. We introduce the following:

$$P_{\text{late}}(t; t_{\text{dead}}, \lambda) = \exp(-\lambda(t - t_{\text{dead}})_+)$$

where $(x)_+ = \max(x, 0)$, t_{dead} represents a characteristic developmental deadline, and λ controls how rapidly contributions lose effectiveness after this point.

2.3 Hemodynamic model

To reiterate, complex hemodynamic models can be incredibly useful. In fact, many have used such models for similar processes including... However, when combining the shear number of different processes, we are faced with an unrealistic computational cost. Thus, as an effort to get around that

barrier, we will use general and macroscopic hemodynamic declarations. In this reduced framework, flow rate is modeled as a time-dependent quantity reflecting increasing cardiac output:

$$Q(t) = Q_0 e^{k_Q t}$$

While geometric changes in the AV canal are incorporated through a dependence on the septation gap.

$$R_{\text{eff}}(t) = R_0 + \kappa_R \; g(t)$$

These quantities are used to define an effective shear stress, which is then mapped to a dimensionless hemodynamic signal that modulates ECM maturation and contributes to resistance against closure.

$$\tau(t) = \frac{4\mu Q(t)}{\pi R_{\text{eff}}(t)^3}$$

This formulation allows us to incorporate the essential feedback between flow, geometry, and tissue development without introducing unnecessary spatial complexity, while preserving the key biological insight that mechanical forces play an active role in determining septation outcomes.

Similar to the approach throughout this mathematical model, we will normalize our variables. Thus, we introduce $N(t)$, a normalized value for our shear stress that follows:

$$N(t) = \frac{\tau(t)}{\tau^*}$$

Where τ^* will be obtained through literature values for normal shear stresses in the AV canal.

2.4 Dynamical system

As with any dynamical system, I will define a system of governing differential equations. To model atrioventricular septation as a time-dependent process, we construct a coupled system dependent on time. Each state variable represents a biological process whose magnitude and timing contribute to septation success. The goal of this system is to capture the effective dynamics of tissue accumulation, remodeling, and structural closure with respect to development.

We adopt a system of stochastic differential equations (SDEs) to represent these dynamics. The deterministic components describe the average behavior of each subsystem based on known biological mechanisms, including cellular influx, proliferation, decay, and mechanical feedback. In addition, stochastic terms are included to account for intrinsic developmental variability and embryo-to-embryo heterogeneity, which are essential for capturing incomplete penetrance.

Each equation is constructed as a balance between processes that promote growth or maturation and processes that oppose or degrade it. Temporal window functions previously defined restrict when specific processes are active, while coupling terms allow interactions between subsystems. In particular, tissue growth depends on both intrinsic biological programs and the surrounding mechanical environment, and structural closure emerges from the combined influence of multiple interacting variables.

This system allows us to represent atrioventricular septation as a coupled, time-dependent system in which success or failure arises from the accumulation of contributions across multiple processes within a finite developmental window.

The evolution of endocardial cushion mesenchymal mass is modeled as a combination of cellular influx, proliferation, and loss:

$$dC = [\alpha_{\text{EMT}} W_{\text{EMT}}(t) + r_C C \phi_A(A) \phi_E(E) - \delta_C C]\, dt + \sigma_C C\, dW_C$$

The first term ($\alpha_{EMT} W_{EMT}(t)$) represents the influx of mesenchymal cells through endocardial-to-mesenchymal transition (EndMT), restricted to a finite developmental window. The second term ($r_C C \phi_A(A) \phi_E(E)$) captures proliferation of cushion cells, modulated by atrioventricular canal competence and ECM state, reflecting the dependence of tissue growth on both signaling and structural environment. The third term ($\delta_C C$) represents cell loss. A stochastic term is included to capture variability in growth dynamics across embryos.

The contribution of DMP tissue is modeled as:

$$dD = [\alpha_{\text{DMP}} W_{\text{DMP}}(t) \phi_A(A) P_{\text{late}}(t) + r_D D - \delta_D D]\, dt + \sigma_D D\, dW_D$$

The first term represents the influx of DMP-derived tissue during a restricted developmental window, modulated by atrioventricular canal competence and reduced by a timing penalty that captures the diminished effectiveness of late contributions. The second and third terms represent proliferation and loss, respectively. As with cushion growth, stochasticity is included to account for variability in developmental trajectories.

The geometric septation gap is modeled as:

$$dG = [-k_G W_{\text{close}}(t)(\omega_C c + \omega_D d)(\omega_E E + \omega_A A) + k_N \chi(N)]\, dt + \sigma_G dW_G$$

This equation represents the balance between processes that promote structural closure and those that oppose it. The first term captures gap reduction driven by the combined contributions of cushion tissue, DMP tissue, ECM maturation, and atrioventricular canal competence, restricted to a closure-specific developmental window. The multiplicative structure reflects the requirement that multiple subsystems must act together to achieve effective closure.

The second term represents resistance to closure arising from hemodynamic forces, modeled as a function of the mechanical environment. The stochastic term captures variability in closure dynamics.

Unlike the other state variables, $G(t)$ represents a geometric quantity and provides a direct structural measure of septation progress. As such, it serves as the primary link between developmental dynamics and anatomical outcome.

ECM maturation is modeled as follows:

$$dE = [k_E(1 - E)\psi_N(N)\psi_C(c) - \gamma_E E]\, dt + \sigma_E dW_E$$

The first term represents ECM deposition and organization, which depends on both hemodynamic stimuli and the presence of sufficient cushion tissue. The factor $(1 - E)$ ensures saturation as maturation approaches completion. The second term represents ECM turnover or degradation. The stochastic component reflects variability in remodeling processes.

Finally, AV canal competence is modeled as:

$$dA = [k_A W_{\text{AVC}}(t)(1 - A) - \gamma_A A]\, dt + \sigma_A dW_A$$

This equation represents the establishment and maintenance of the inductive signaling environment required for proper septation. Competence is activated within a developmental window and saturates at a maximal level, while decay captures loss or disruption of signaling. This variable serves as a coarse-grained representation of multiple underlying molecular pathways.

2.5 Activation functions

Several processes in the model depend on the current state of other variables and are therefore modulated using nonlinear activation functions. These functions are chosen to represent thresholded and saturating biological responses while remaining smooth and bounded.

We define the following state-dependent activation terms:

$$\phi_A(A) = \frac{A^{n_A}}{A^{n_A} + K_A^{n_A}}, \quad \phi_E(E) = \frac{E^{n_E}}{E^{n_E} + K_E^{n_E}}$$

$$\psi_N(N) = \frac{N^{n_N}}{N^{n_N} + K_N^{n_N}}, \quad \psi_C(c) = \frac{c^{n_C}}{c^{n_C} + K_C^{n_C}}$$

$$\chi(N) = (N - N_{\text{crit}})_+$$

The functions $\phi_A(A)$ and $\phi_E(E)$ modulate cushion proliferation by atrioventricular canal competence and ECM maturation, respectively. These terms ensure that when A or E are low, their influence on growth is minimal, while at higher values their effect saturates. This reflects the requirement that sufficient signaling competence and matrix organization must be established before effective tissue expansion can occur.

Similarly, $\psi_H(H)$ and $\psi_C(c)$ regulate ECM maturation as a function of hemodynamic stimuli and available cushion tissue. These functions encode the observation that both mechanical forces and structural substrate must exceed minimal levels before contributing meaningfully to matrix remodeling.

The function $\chi(H)$ represents a thresholded hemodynamic resistance to closure. Shear stress below a critical level has negligible opposing effect, while stresses above this threshold increasingly resist septation. This captures the idea that mechanical forces act as a nonlinear barrier to closure rather than a purely linear load.

All activation functions are formulated as Hill-type relationships, where K defines the half-activation level and n controls the sharpness of the transition. This choice provides a smooth approximation of cooperative or threshold-like biological behavior while maintaining numerical stability and interpretability.

2.6 Genotype to parameter mapping

To connect genetic variation to developmental dynamics, we model each parameter θ_j as a function of genotype. Rather than assuming a single causal gene, we represent genetic effects as shifts in biologically interpretable model parameters.

We define this relationship in log-space as

$$\log \theta_j = \log \theta_j^{(0)} + \beta_j^{\text{T21}} + \sum_{m=1}^{M} B_{jm} z_m + \epsilon_j$$

Here, $\theta_j^{(0)}$ denotes the baseline value of parameter j under typical (euploid) development. The term β_j^{T21} represents the systematic shift associated with trisomy 21, capturing global dosage effects such as altered signaling, extracellular matrix composition, or cellular proliferation rates.

The summation term $\sum_{m=1}^{M} B_{jm} z_m$ represents the contribution of additional genetic modifiers. Each z_m encodes the presence or magnitude of a variant or pathway-level perturbation, while B_{jm} quantifies how strongly that modifier influences parameter j. This structure allows multiple small genetic effects to combine and influence developmental processes in a distributed manner.

The stochastic term ϵ_j captures unmodeled variability, including epigenetic differences, environmental influences, and intrinsic biological noise. This component is essential for representing incomplete penetrance, as embryos with similar genetic profiles may still exhibit different developmental trajectories.

Modeling parameters in log-space ensures that all θ_j remain positive and allows genetic effects to act multiplicatively on biological processes. Under this formulation, trisomy 21 does not directly determine the phenotype; instead, it shifts the parameter landscape such that the developmental system is closer to a failure threshold, making it more susceptible to additional perturbations.

This mapping provides a flexible framework for linking genotype, developmental dynamics, and phenotypic outcome within a unified model.

2.7 Linear and nonlinear phenotype maps

To relate developmental state to phenotypic outcome, we define a scalar measure of septation sufficiency evaluated at a terminal developmental time T. This quantity integrates contributions from multiple subsystems, including cushion mass, DMP contribution, ECM maturation, and atrioventricular canal competence.

We first consider a linear (additive) formulation:

$$S_{\text{lin}}(T) = w_C c(T) + w_D d(T) + w_E E(T) + w_A A(T)$$

In this representation, each component contributes independently to overall septation success, with weights w_i reflecting their relative importance. This formulation corresponds to a model in which deficits in different subsystems accumulate additively, and partial compensation between processes is possible.

To test whether developmental failure arises from interactions between subsystems, we also define a nonlinear (multiplicative) formulation:

$$S_{\text{nonlin}}(T) = c(T)^{w_C} d(T)^{w_D} E(T)^{w_E} A(T)^{w_A}$$

Taking the logarithm yields

$$\log S_{\text{nonlin}}(T) = w_C \log c(T) + w_D \log d(T) + w_E \log E(T) + w_A \log A(T)$$

In this formulation, successful septation requires that all components reach sufficient levels, as low values in any variable disproportionately reduce the overall score. This captures a cooperative or bottleneck-like behavior in which multiple subsystems must function simultaneously for closure to occur.

Comparing these two formulations allows us to distinguish between additive and interaction-driven models of developmental failure. If the linear model is sufficient, AVSD risk arises from independently accumulating deficits. If the nonlinear model better predicts outcomes, this supports a mechanism in which coupled failures across multiple processes drive septation failure.

In both cases, the scalar score $S(T)$ is evaluated against a threshold to determine phenotypic outcome, consistent with a threshold-based model of AVSD.

3 Parameterization

3.1 Literature-derived anchors

The model parameters are chosen to reflect biologically interpretable processes governing atrioventricular septation. Rather than treating parameters as abstract coefficients, each parameter

corresponds to a specific mechanism, including cellular influx, proliferation, extracellular matrix remodeling, mechanical feedback, and developmental timing. For clarity, parameters are grouped according to the processes they regulate.

Developmental processes are restricted to finite temporal windows that correspond to known stages of embryonic heart formation. Parameters defining these windows include onset times (t_{on}), offset times (t_{off}), and transition widths (Δ), which together determine when a process is active and how sharply it turns on or off.

These parameters reflect the staged nature of development. For example, EndMT occurs early, DMP contribution follows within a narrow interval, and structural fusion occurs later. Additional parameters, such as t_{dead} and λ, define timing penalties that reduce the effectiveness of processes occurring after critical developmental deadlines. Together, these parameters enforce the principle that both timing and magnitude are required for successful septation.

Cellular processes are governed by parameters that describe influx, proliferation, and loss. The parameter α_{EMT} represents the rate of mesenchymal cell influx into the atrioventricular cushions through EndMT, while analogous terms govern DMP contribution. Proliferation rates, such as r_C and r_D, determine how rapidly tissue mass expands once established, and decay parameters (e.g., δ_C, δ_D) represent cell loss or turnover.

These parameters collectively determine the magnitude of tissue contributions to septation. They encode both the supply of new cells and the ability of existing tissue to expand, capturing the balance between growth and loss that governs structural development.

Extracellular matrix remodeling and developmental signaling are controlled by parameters such as k_E and k_A, which govern the rates of ECM maturation and atrioventricular canal competence, respectively. Corresponding decay parameters (e.g., γ_E, γ_A) represent turnover or loss of these states.

These parameters are distinct from purely cellular terms, as they regulate the quality of the developmental environment rather than the quantity of tissue. ECM maturation influences tissue stiffness and organization, while AV canal competence reflects the presence of appropriate signaling conditions. Together, they modulate how effectively cellular contributions translate into structural closure.

3.2 Parameter ranges and priors

Parameter values were not selected arbitrarily, but instead constrained using physiological measurements, developmental timing data, and known scaling relationships from cardiovascular and embryologic literature. Given the absence of direct quantitative measurements for many quantities in atrioventricular septation, ranges were chosen by combining (i) experimentally measured physical quantities, (ii) biologically observed timescales, and (iii) consistency with known developmental behavior.

The proliferation parameter r_C represents the effective growth rate of mesenchymal cushion cells following EndMT. During early cardiac development, EndMT generates a rapidly expanding mesenchymal population that fills the cardiac jelly. Experimental studies show that after initial EMT induction, cushion growth is dominated by proliferation rather than continued influx.

Typical cellular proliferation timescales in embryonic tissues range on the order of hours to a day, implying dimensionless growth rates on the order of unity when time is measured in days. Accordingly, r_C is chosen to reflect biologically plausible doubling behavior over developmental timescales rather than instantaneous expansion.

The influx parameter α_{EMT} is derived from the fact that EndMT is a transient but intense process, occurring over a narrow developmental window (approximately E9.5–E10.5 in mouse).

During this period, a large number of endothelial cells transition into mesenchymal cells and invade the cardiac jelly. Studies demonstrate that shear stress and signaling pathways such as TGF-β regulate the magnitude of this influx, with increased hemodynamic load enhancing mesenchymal cell density.

Because EndMT is temporally concentrated but produces a substantial cell population, α_{EMT} is chosen to be large relative to proliferation terms, representing a burst-like influx rather than a continuous process.

The parameter k_E represents the effective rate of extracellular matrix (ECM) maturation, including deposition, organization, and crosslinking. ECM development occurs over a longer timescale than initial cellular influx, as the cardiac jelly transitions into structured valve tissue.

Experimental observations indicate that ECM remodeling is regulated both by biochemical signaling and mechanical forces, with shear stress and cyclic strain influencing matrix organization and alignment. Unlike rapid cellular influx, ECM maturation is gradual and saturating, motivating a rate parameter that is slower and bounded relative to proliferation.

Thus, k_E is selected to reflect a process that evolves over multiple developmental days rather than rapidly equilibrating.

Hemodynamic parameters are among the most directly constrained by physical measurements. Blood viscosity μ and density ρ are taken from well-established physiological values for mammalian blood. Shear stress scales as

$$\tau \sim \frac{\mu Q}{R^3},$$

so these parameters directly determine the magnitude of mechanical forces experienced by developing tissues.

Experimental studies report that endothelial cells experience shear stresses spanning a wide physiological range, with values in the order of several Pascals depending on vessel size and flow conditions. Additionally, shear stress is known to regulate EndMT and endothelial signaling in regions such as the AV canal.

These observations constrain the choice of μ and ρ such that resulting shear stress magnitudes fall within biologically realistic ranges, ensuring that mechanotransduction effects are neither negligible nor unrealistically dominant.

The parameter k_G represents the effective rate at which tissue contributions translate into reduction of the septation gap. Unlike cellular or mechanical parameters, this quantity is not directly measurable and instead represents an aggregate of multiple processes, including tissue growth, ECM remodeling, and mechanical opposition.

To constrain k_G, we use developmental timing. In mouse models, major cushion fusion and septation occur over a finite interval (approximately E10.5–E12.5). Therefore, k_G must be large enough to allow substantial gap reduction within this window, but not so large as to produce instantaneous closure.

This leads to a range consistent with gradual but decisive structural change over developmental timescales.

All parameter ranges were additionally constrained by requiring that the model reproduce known qualitative behaviors of AV septation, including: gradual accumulation of cushion mass followed by remodeling, sensitivity of development to hemodynamic perturbations, inability of late-stage compensation to fully rescue early deficits, and realistic timing of septation completion.

Rather than representing precise measurements, these ranges define a biologically consistent parameter space within which the coupled dynamical system produces physiologically plausible trajectories. Representative parameter ranges consistent with these constraints are summarized below in Table 2.

Parameter	Description	Type	Range
r_C	Cushion cell proliferation rate	Kinetic	0.3–1.0
α_{EMT}	EndMT-driven mesenchymal influx	Source term	500–5000
k_E	ECM maturation rate constant	Kinetic	0.2–2.0
k_G	Effective septation (closure) rate	Geometric/kinetic	0.2–3.0
μ	Blood dynamic viscosity	Physical	0.003 Pa · s
ρ	Blood density	Physical	1060 kg/m^3

Table 2: Summary of key model parameters, including biological, mechanical, and effective kinetic terms used in the AVSD dynamical system.

3.3 Genetic axes

To represent genetic variation in a biologically interpretable manner, we introduce a set of latent genetic axes $\{z_m\}_{m=1}^{M}$ that capture distinct modes of perturbation relevant to atrioventricular septation. Rather than modeling individual genes explicitly, these axes represent pathway-level or functional effects that influence developmental processes.

Each axis z_m corresponds to a coherent biological mechanism that can affect one or more parameters in the dynamical system. Examples of such axes include alterations in cell adhesion, extracellular matrix composition, signaling pathway activity, and mechanotransduction sensitivity. This abstraction reflects the observation that congenital heart defects, particularly in trisomy 21, are not typically driven by single high-effect variants but instead arise from distributed perturbations across multiple pathways.

The mapping coefficients B_{jm} determine how strongly each genetic axis influences a given parameter θ_j. This structure allows a single biological perturbation to affect multiple aspects of development simultaneously. For example, an axis representing impaired extracellular matrix regulation may influence both ECM maturation rates and tissue mechanical properties, while an axis related to signaling competence may affect both EndMT efficiency and atrioventricular canal competence.

Importantly, the axes are not assumed to be independent at the molecular level, but are treated as independent components in the model for tractability. This provides a low-dimensional representation of genetic variability while preserving the ability to capture coordinated effects across multiple developmental processes.

This formulation allows genetic variation to be represented as shifts in a small number of biologically meaningful directions in parameter space, rather than as a high-dimensional set of individual gene effects. As a result, the model can capture how combinations of small perturbations interact to influence developmental trajectories and determine whether the system reaches the threshold required for successful septation.

In the context of trisomy 21, these axes can be interpreted as reflecting dosage-sensitive pathways known to be perturbed in Down syndrome, such as cell adhesion (e.g., DSCAM-related effects), ECM regulation, and signaling pathways involved in cardiac development. This provides a natural link between genotype, pathway-level dysfunction, and emergent developmental failure.

4 Simulation Framework

4.1 Numerical methods

The developmental model was simulated as a coupled system of stochastic differential equations (SDEs) over the state vector

$$\mathbf{u}(t) = \big[C(t),\, D(t),\, G(t),\, E(t),\, A(t)\big]^\top,$$

where $C(t)$ denotes endocardial cushion mesenchymal mass, $D(t)$ denotes dorsal mesenchymal protrusion (DMP) contribution, $G(t)$ denotes the remaining geometric septation gap, $E(t)$ denotes extracellular matrix (ECM) maturation, and $A(t)$ denotes atrioventricular canal competence. The deterministic part of the system captures the mean developmental trajectory implied by the biological mechanisms introduced above, while the stochastic part represents embryo-to-embryo variability and intrinsic developmental noise.

Because the state variables obey biologically meaningful bounds, we did not integrate the original variables directly. In particular, the extensive variables

$$C(t) > 0, \qquad D(t) > 0, \qquad G(t) > 0$$

must remain strictly positive, whereas the intensive variables

$$0 < E(t) < 1, \qquad 0 < A(t) < 1$$

are naturally bounded on the unit interval. Direct numerical integration of the original SDE system can violate these constraints, especially in the presence of multiplicative noise. To enforce the admissible state space, we introduced transformed variables

$$x_C = \log C, \qquad x_D = \log D, \qquad x_G = \log G,$$

for the positive extensive quantities, and

$$x_E = \operatorname{logit}(E) = \log\left(\frac{E}{1-E}\right), \qquad x_A = \operatorname{logit}(A) = \log\left(\frac{A}{1-A}\right),$$

for the bounded intensive variables. The original states are recovered by

$$C = e^{x_C}, \qquad D = e^{x_D}, \qquad G = e^{x_G},$$

and

$$E = \frac{1}{1+e^{-x_E}}, \qquad A = \frac{1}{1+e^{-x_A}}.$$

This transformation guarantees positivity of C, D, and G, and ensures that E and A remain in $(0,1)$ for all simulated times, eliminating the need for ad hoc post-step clipping.

Let the original model in physical variables be written abstractly as

$$d\mathbf{u} = \mathbf{f}(\mathbf{u}, t; \boldsymbol{\theta})\, dt + \boldsymbol{\Sigma}(\mathbf{u}, t; \boldsymbol{\theta})\, d\mathbf{W}(t),$$

where $\mathbf{f}$ is the drift vector, $\boldsymbol{\Sigma}$ is the diffusion matrix, $\boldsymbol{\theta}$ denotes the parameter set, and

$$\mathbf{W}(t) = \big[W_C(t),\, W_D(t),\, W_G(t),\, W_E(t),\, W_A(t)\big]^\top$$

is a vector of independent Wiener processes. In the present work, independent noise processes were assigned to each developmental subsystem in order to represent state-specific fluctuations without imposing an assumed correlation structure between subsystems. This choice provides the most interpretable initial representation of stochastic variability, while leaving correlated noise models as a natural extension for future work.

The transformed system was then written in terms of the variable

$$\mathbf{x}(t) = \big[x_C(t),\, x_D(t),\, x_G(t),\, x_E(t),\, x_A(t)\big]^\top.$$

For the log-transformed variables associated with positive states, Itô's lemma gives, for example,

$$dx_C = \left[\frac{1}{C} f_C(\mathbf{u}, t) - \frac{1}{2C^2} g_C(\mathbf{u}, t)^2\right] dt + \frac{1}{C} g_C(\mathbf{u}, t)\, dW_C,$$

with analogous expressions for x_D and x_G. Here f_C and g_C denote the drift and diffusion terms associated with the C equation in the original system. Similarly, for the logit-transformed bounded variables,

$$x_E = \log\left(\frac{E}{1-E}\right),$$

Itô's lemma yields

$$dx_E = \left[\frac{1}{E(1-E)} f_E(\mathbf{u}, t) + \frac{2E-1}{2E^2(1-E)^2} g_E(\mathbf{u}, t)^2\right] dt + \frac{1}{E(1-E)} g_E(\mathbf{u}, t)\, dW_E,$$

with an analogous expression for x_A. These transformed equations were integrated numerically, while all biological interactions, hemodynamic feedback, and phenotype evaluation were computed using the recovered physical variables (C, D, G, E, A).

The deterministic drift terms were defined by the developmental mechanisms introduced in Section 3. In compact form, the system can be represented as

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u}, t; \boldsymbol{\theta}),$$

where each component depends on time-window functions, nonlinear activation functions, and hemodynamic feedback. For example, the effective shear stress was evaluated as

$$\tau(t) = \frac{4\mu Q(t)}{\pi R_{\text{eff}}(t)^3}, \qquad Q(t) = Q_0 e^{k_Q t}, \qquad R_{\text{eff}}(t) = R_0 + \kappa_R g(t),$$

with

$$g(t) = \frac{G(t)}{G^*}, \qquad N(t) = \frac{\tau(t)}{\tau^*}.$$

These quantities enter the drift terms through nonlinear activation functions such as

$$\psi_N(N) = \frac{N^{n_N}}{N^{n_N} + K_N^{n_N}}, \qquad \chi(N) = (N - N_{\text{crit}})_+,$$

and through smooth developmental timing windows

$$W(t; t_{\text{on}}, t_{\text{off}}, \Delta) = \sigma\left(\frac{t - t_{\text{on}}}{\Delta}\right) \sigma\left(\frac{t_{\text{off}} - t}{\Delta}\right), \qquad \sigma(x) = \frac{1}{1 + e^{-x}}.$$

The use of smooth windowing functions avoids introducing hard discontinuities into the right-hand side of the system, which improves numerical stability while still preserving the biologically restricted timing of individual developmental processes.

Simulation proceeded in two stages. First, the deterministic system

$$\frac{d\mathbf{x}}{dt} = \tilde{\mathbf{f}}(\mathbf{x}, t; \boldsymbol{\theta})$$

was solved in the absence of noise in order to verify baseline developmental trajectories, assess stiffness, and confirm that the chosen parameter sets reproduced qualitatively plausible temporal behavior. Because the model combines fast cellular influx, slower remodeling dynamics, sharp but smooth gating functions, and nonlinear feedback terms, the system exhibits disparate timescales and may be numerically stiff. Deterministic baseline simulations were therefore solved using stiff ODE integrators, such as implicit Runge–Kutta or backward differentiation formula (BDF) methods.

After deterministic validation, stochastic simulations were performed by solving the transformed Itô SDE system

$$d\mathbf{x} = \tilde{\mathbf{f}}(\mathbf{x}, t; \boldsymbol{\theta})\, dt + \tilde{\mathbf{G}}(\mathbf{x}, t; \boldsymbol{\theta})\, d\mathbf{W}(t),$$

where $\tilde{\mathbf{f}}$ and $\tilde{\mathbf{G}}$ denote the transformed drift and diffusion terms. Numerical integration was carried out using strong-order stochastic Runge–Kutta methods implemented in `DifferentialEquations.jl`. This choice was made because the model contains multiplicative noise, nonlinear state dependence, and time-localized activation functions, all of which require accurate handling of both drift and diffusion to avoid biasing the developmental trajectories. Independent Wiener increments were generated for each state variable at each time step, with noise amplitudes determined by the corresponding diffusion coefficients.

The simulation interval was chosen to span the relevant developmental window from approximately embryonic day E9.0 to E12.5. Letting $t = 0$ correspond to E9.0, trajectories were evolved on

$$t \in [0, T], \qquad T \approx 3.5 \text{ days},$$

with temporal resolution sufficient to resolve EndMT initiation, DMP contribution, ECM maturation, and gap closure dynamics. Adaptive time-stepping was used where appropriate to control local truncation error and to accommodate rapid changes near activation boundaries. In stochastic simulations, step sizes were also required to remain sufficiently small to accurately resolve multiplicative noise in the transformed coordinates.

For each simulated embryo, the computational pipeline consisted of: (i) constructing a parameter set $\boldsymbol{\theta}$ from the baseline model and genotype-dependent perturbations, (ii) solving the transformed SDE system over the developmental interval, (iii) reconstructing the physical state variables from the transformed coordinates, and (iv) evaluating terminal structural and functional outcome measures at time T. Terminal outcomes included the residual gap $G(T)$, normalized tissue states $c(T)$ and $d(T)$, and scalar sufficiency scores

$$S_{\text{lin}}(T) = w_C c(T) + w_D d(T) + w_E E(T) + w_A A(T),$$

and

$$S_{\text{nonlin}}(T) = c(T)^{w_C} d(T)^{w_D} E(T)^{w_E} A(T)^{w_A}.$$

Large-scale ensemble simulations were used to estimate prevalence and characterize developmental variability across populations. For a cohort of N embryos, the model was solved independently for each sampled realization,

$$\{\boldsymbol{\theta}^{(i)}, \mathbf{W}^{(i)}(t)\}_{i=1}^{N},$$

and phenotype statistics were computed from the resulting terminal states. Because ensemble simulation is embarrassingly parallel, independent trajectories were executed in parallel to enable large population studies without altering the mathematical formulation of the model.

Several numerical validation checks were performed. First, deterministic baseline trajectories were required to exhibit biologically plausible qualitative behavior, including growth of cushion and DMP contributions, progressive ECM maturation, and monotonic reduction of the septation gap under normal development. Second, transformed-state integration was verified to preserve admissible domains under all tested parameter sets. Third, stochastic simulations were compared against deterministic baselines to ensure that the mean stochastic trajectory remained consistent with the corresponding noise-free solution when diffusion amplitudes were small. Finally, convergence with respect to time-step tolerance and ensemble size was assessed to confirm that predicted prevalence and sensitivity estimates were not numerical artifacts.

All simulations were implemented in Julia using the `DifferentialEquations.jl` ecosystem, which provides a unified framework for stiff ODE solving, Itô SDE integration, and large-scale ensemble simulation. This computational framework allowed the model to capture nonlinear developmental dynamics, state-dependent stochasticity, and population-level variability within a single numerically consistent formulation.

4.2 Population sampling

To model incomplete penetrance and inter-individual variability, population-level simulations were performed by sampling over genetic and stochastic sources of variation. Each simulated embryo was characterized by a genotype profile consisting of a trisomy indicator and a vector of latent modifier variables,

$$\mathbf{z} = (z_1, z_2, \ldots, z_M),$$

representing pathway-level perturbations such as altered cell adhesion, extracellular matrix regulation, or signaling competence.

For each embryo, model parameters were generated by mapping the genotype profile into parameter space according to the log-linear relationship described in Section 3.X. Latent modifier variables were sampled from prescribed prior distributions (e.g., Gaussian or bounded uniform), while stochastic variability in developmental processes was introduced through independent Wiener processes in the SDE formulation.

Population cohorts were constructed for both euploid and trisomy 21 conditions. For each cohort, a large number of embryos ($N \sim 10^3$–10^5) were simulated, with each realization involving:

1. Sampling of latent genetic modifiers,
2. Mapping to a full parameter set,
3. Simulation of the developmental trajectory via the SDE system,
4. Evaluation of terminal developmental state and phenotype.

Phenotypic outcomes were determined using both geometric and functional criteria, including the residual septation gap $G(T)$ and scalar sufficiency measures $S(T)$ derived from linear and nonlinear phenotype maps. This framework enabled estimation of phenotype prevalence, distribution of developmental trajectories, and the influence of genetic perturbations across populations.

Sampling strategies were chosen to ensure adequate coverage of parameter space, including the use of stratified or quasi-random sampling (e.g., Latin hypercube or Sobol sequences) where

appropriate. Parallel ensemble simulation was used to efficiently generate large populations and compute robust statistics.

4.3 Calibration targets

Model calibration was performed to ensure consistency with known biological and epidemiological observations. Rather than fitting individual parameters directly to sparse measurements, calibration was conducted at the level of emergent system behavior.

Primary calibration targets included Phenotype prevalence, Developmental timing, Trajectory plausibility, and Mechanical consistency. The model was constrained to reproduce the observed prevalence of atrioventricular septal defects in trisomy 21 populations (approximately 15–33%), as well as substantially lower prevalence in euploid controls. Additionally, simulated trajectories were required to align with known developmental windows, including initiation of endocardial-to-mesenchymal transition (E9.5), dorsal mesenchymal protrusion contribution (E9.5–E10.5), and septation completion (approximately E12.5). State variables were required to exhibit biologically consistent dynamics, including monotonic or saturating growth of cushion mass, progressive ECM maturation, and gradual reduction of the septation gap. Moreover, hemodynamic quantities such as shear stress were constrained to remain within physiologically realistic ranges derived from cardiovascular literature.

Calibration was performed within a Bayesian framework, treating model parameters as random variables with prior distributions informed by biological knowledge and literature-derived ranges. Posterior distributions were inferred by comparing simulated outputs to calibration targets using likelihood-based or approximate Bayesian methods.

This approach allows uncertainty in parameter values to be propagated through the model, enabling probabilistic prediction of developmental outcomes and providing a principled framework for quantifying the contributions of genetic, mechanical, and stochastic factors to AVSD risk.

5 Results

5.1 Baseline model behavior

5.2 Failure boundary structure

5.3 Linear vs nonlinear comparison

5.4 Sensitivity analysis

5.5 Effect of genetic perturbations

6 Discussion

6.1 Interpretation

6.2 Mechanistic insights

6.3 Comparison to existing literature

6.4 Limitations

- Parameter uncertainty
- Mouse vs human translation

- Simplified hemodynamics
- No direct experimental validation

6.5 Future directions

- Integration with scRNA data
- Patient-specific modeling
- Experimental validation

7 Conclusion

We present a mechanistically grounded, genetically informed dynamical model of AV septal development that formalizes AVSD as a threshold-crossing process. This framework provides a foundation for linking genetic perturbations to morphogenetic outcomes and enables quantitative exploration of developmental failure in CHD.

Data and Code Availability

Acknowledgments

References